

Breaking the Vision–Action Shortcut: Latent Interface Training for Generalizable Robotics Foundation Models

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Abstract—Robot foundation models achieve strong in-distribution performance but often degrade under visual distribution shifts. When learning to generate actions from pretrained visual representations, models may exploit task-irrelevant visual cues that correlate with demonstrated actions within the training distribution. Such *vision–action shortcuts* can undermine generalization when these correlations change under distribution shifts. Mitigating these shortcuts requires constraining how visual information is used for action generation while preserving task-relevant spatial information. We propose Latent Interface Training (LIT), a framework-agnostic two-stage strategy that first establishes a spatial-goal-conditioned action prior without images, then constrains visual conditioning through a pose-supervised latent interface. Stage 1 trains the action expert to generate action chunks conditioned on language, robot state, and each demonstrated chunk’s terminal SE(3) end-effector pose, learning goal-directed action generation independently of visual cues. Stage 2 introduces a latent interface that aggregates visual and semantic representations and serves as the pretrained action expert’s only visual conditioning pathway. The interface is supervised to reconstruct the terminal pose previously used to condition Stage 1, encouraging it to retain the goal-relevant spatial information needed for action generation. Across four vision–language–action and world–action architectures— $\pi_{0.5}$, MolmoAct2, FAST-WAM, and ImageWAM—LIT improves overall LIBERO-Plus success by 3.87–10.70 percentage points while preserving or improving average LIBERO success. Real-world evaluations show 13.30–16.70 percentage-point gains in success aggregated across three tasks under unseen camera configurations, lighting variations, and distractors. Project page: magiclab-nus.github.io/LIT

I. INTRODUCTION

Recent robot foundation models, including representative vision–language–action (VLA) and world–action model (WAM) architectures, combine pretrained vision–language or video backbones with action experts, achieving strong in-distribution manipulation performance [1], [2], [3], [4]. Visual representations condition action generation alongside language and robot-state information [5], [6], as illustrated in Fig. 1 (left). Robust generalization under visual distribution shifts, however, remains a challenge [7].

Under visual conditioning, action experts may learn *vision–action shortcuts*: reliance on task-irrelevant visual cues that correlate with demonstrated actions during training but become unreliable under distribution shifts [7], [8]. Limited visual diversity in robot demonstrations can further encourage this reliance, undermining generalization to changes in viewpoint, appearance, or sensing conditions. Yet visual representations also encode task-relevant spatial information essential for action generation. Mitigating these shortcuts therefore requires reducing sensitivity to task-irrelevant visual variations while preserving responsiveness to task-relevant changes that require different actions.

Existing efforts improve generalization through representation enhancement and stage-wise action pretraining. Representation-enhanced methods introduce structured spatial or motion information to ground action generation in task-relevant geometry [9], [10], [11]. However, enriching the available information does not directly constrain how the action expert uses it, leaving room for vision–action shortcuts. Stage-wise approaches first learn language-conditioned action priors without images, then use them to initialize visual policy training [12], [13]. However, these approaches lack an explicit spatial goal for each action chunk during pretraining, and subsequent training introduces visual conditioning without explicitly constraining its use. Image-free pretraining alone therefore leaves the policy susceptible to shortcuts once visual conditioning is introduced. These limitations motivate a central question: *How can action learning and visual conditioning be structured to mitigate vision–action shortcuts while preserving the task-relevant spatial information needed for action generation?*

To address this challenge, we propose Latent Interface Training (LIT), a model-agnostic two-stage strategy that first learns a spatial-goal-conditioned action prior without images, then introduces visual conditioning through a pose-supervised latent interface (Fig. 1, right). **In the first stage**, the action expert is conditioned on language and robot-state representations only from a frozen pretrained backbone, together with an encoding of each demonstrated action chunk’s terminal SE(3) end-effector pose. This establishes a spatial-goal-conditioned action prior without visual inputs. **In the second stage**, learnable latent tokens cross-attend to visual and semantic backbone representations and provide layer-wise conditioning to the pretrained action expert, serving as its only visual conditioning pathway. A pose-reconstruction objective supervises these tokens to recover the same terminal pose used in Stage 1, encouraging them to retain goal-relevant spatial information. The shared spatial target thus

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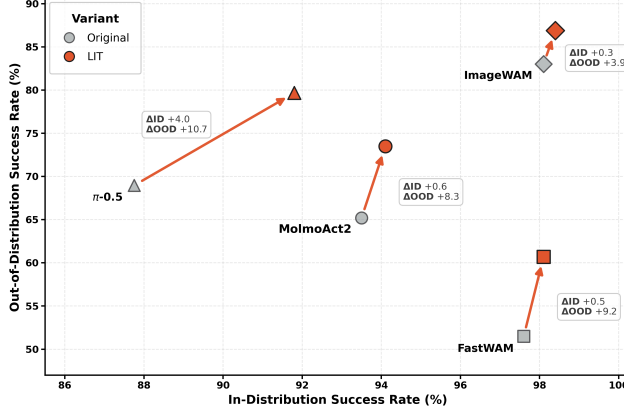
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VLA/WAM Training



Performance improvements in-distribution/out-of-distribution



Latent Interface Training

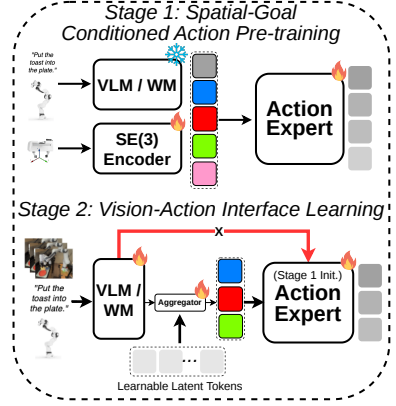


Fig. 1. **Training paradigms and generalization performance.** Standard training conditions the action expert on visual representations (left). LIT (right) first learns a spatial-goal-conditioned action prior without images to avoid relying on visual correlations during pre-training (Stage 1), then introduces visual conditioning through a pose-supervised latent interface (Stage 2). Across four VLA and WAM architectures, LIT improves out-of-distribution success rates while preserving or improving in-distribution success rates for LIBERO and LIBERO-Plus (center).

connects action-prior learning with visual interface learning, while the exclusive interface constrains visual conditioning to mitigate vision-action shortcuts.

Our contributions are threefold. First, we introduce **Latent Interface Training (LIT)**, a model-agnostic training strategy for improving generalization in robot foundation models. LIT combines image-free, spatial-goal-conditioned action pretraining with a pose-supervised latent interface that serves as the action expert’s only visual conditioning pathway. Second, across $\pi_{0.5}$ [1], MolmoAct2 [2], FAST-WAM [3], and ImageWAM [4], LIT improves overall LIBERO-Plus [7] success by 3.87%–10.70% while preserving or improving average LIBERO [14] success. These results, together with ablations and counterfactual analyses, support LIT’s effectiveness in mitigating vision-action shortcuts while preserving task-relevant spatial information. Third, real-world evaluations demonstrate LIT’s robustness to unseen camera configurations, lighting variations, and distractors, with gains of 13.3–16.7 percentage points in success aggregated across three manipulation tasks.

II. RELATED WORK

We review how visual information conditions action generation in robot foundation models, followed by representation design for generalizable control and action pretraining for downstream policy learning.

A. Visual Conditioning in Robot Foundation Models

Large-scale cross-embodiment datasets and generalist policies have expanded the scope of transferable robot control [15], [16]. Within robot foundation models, visual information enters action generation through different architectural pathways. Autoregressive VLAs, such as RT-2 [17] and OpenVLA [18], predict discretized action tokens from a shared visual-language context. Diffusion Policy [5] generates action sequences through conditional denoising, while CogACT [19] conditions a specialized diffusion action

module on VLM representations. For flow-based action generation, π_0 [6] and $\pi_{0.5}$ [1] adopt a mixture-of-transformers design, where the backbone and action expert use separate parameters and interact through joint attention over their tokens. MolmoAct2 [2] instead uses layer-wise cross-attention to condition its action expert on projected key-value representations from corresponding backbone layers. WAMs, including DreamZero [20], FAST-WAM [3], and ImageWAM [4], incorporate representations learned through video or world modeling into action generation.

These architectures commonly expose action-generation components to rich visual representations. Knowledge insulation [21] improves training and knowledge transfer by blocking gradients from the action expert into the backbone, while retaining backbone representations as conditioning inputs to the action expert. Learning actions from these representations may encourage reliance on scene-specific correlations, particularly when training demonstrations exhibit limited visual diversity. This is consistent with shortcut learning and causal confusion in imitation learning [8], [22]. Evaluation studies further document sensitivity to visual shifts and broader generalization challenges in language-conditioned manipulation [7], [23], [24].

B. Representations for Generalizable Robot Control

Prior work improves policy generalization by enriching the information used for action generation. R3M [25] and MVP [26] learn transferable visual features through large-scale video or image pretraining. Other methods introduce spatial, temporal, or semantic structure. TraceVLA [11] augments observations with historical motion traces, while SpatialVLA [9] incorporates egocentric 3D position information. ECoT [27] generates textual reasoning about plans, subtasks, and visually grounded features before predicting actions. MolmoAct [10] structures action prediction through depth tokens and visual reasoning traces, while CoT-VLA [28] predicts future images as visual subgoals. Hierarchical ap-

proaches such as HAMSTER [29] and 3D HAMSTER [30] guide low-level control with predicted 2D and 3D end-effector trajectories, respectively. Crossway Diffusion [31] instead improves control representations through auxiliary state reconstruction.

Another line of work improves robustness by filtering task-irrelevant information. OREO [32] uses object-aware regularization to discourage reliance on nuisance visual cues, while Selective Visual Representations [33] learns a task-conditioned codebook bottleneck to filter visual features. However, richer representations, explicit guidance, and information filtering do not by themselves ensure that visual conditioning retains task-relevant spatial information without exploiting shortcut-prone cues. LIT targets this challenge through a pose-supervised latent interface as the action expert’s only visual conditioning pathway.

C. Action Expert Pretraining for Generalizable Policies

A related line of work pretrains action-generation components to learn reusable priors for subsequent policy learning. Qwen-VLA [13] pretrains a flow-matching decoder conditioned on language and embodiment prompts without images, then introduces visual conditioning. LA4VLA [12] learns action priors from atomic instructions and robot states without images; its sequential LA-to-VLA variant initializes subsequent visual policy training with these priors. Action prior learning [34] uses state-action trajectories without images or language, transferring the learned prior through decoder reuse and early-stage latent distillation. APT [35] instead pretrains a vision-action expert before introducing language conditioning to improve generalization to unseen instructions.

These studies show that action pretraining can facilitate downstream policy learning and improve generalization in different settings. However, pretraining alone does not explicitly constrain how visual information is used during subsequent policy learning, leaving action generation susceptible to scene-specific visual correlations. LIT uses each action chunk’s terminal pose to guide image-free action pretraining, then introduces visual conditioning through a latent interface supervised to reconstruct the same pose. This interface serves as the action expert’s only visual conditioning pathway.

III. METHOD

Figure 2 provides an overview of **Latent Interface Training (LIT)**, a model-agnostic two-stage strategy for reducing vision-action shortcuts while preserving task-relevant spatial information. LIT first establishes an action prior without images, then introduces visual conditioning exclusively through a pose-supervised latent interface. We first formulate the problem, then describe spatial-goal-conditioned action pretraining (Sec. III-B) and latent interface learning (Sec. III-C), followed by framework integration and inference (Sec. III-D).

A. Problem Setup

We consider robot foundation models that combine a pre-trained backbone with an embodiment-specific action expert.

At timestep t , the policy receives a visual observation $\mathbf{o}_t$, a language instruction l , and a robot state $\mathbf{s}_t$, and predicts an action chunk of horizon H :

$$\mathbf{A}_t = (\mathbf{a}_t, \dots, \mathbf{a}_{t+H-1}). \quad (1)$$

For each demonstrated chunk, we use the terminal robot state as the goal:

$$\mathbf{g}_t = [\mathbf{p}_{t+H}; \mathbf{r}_{t+H}; \mathbf{q}_{t+H}] \in \mathbb{R}^8, \quad (2)$$

where $\mathbf{p} \in \mathbb{R}^3$ and $\mathbf{r} \in \mathbb{R}^3$ represent the world-frame end-effector position and axis-angle orientation, and $\mathbf{q} \in \mathbb{R}^2$ contains the gripper joint positions. This goal serves as a conditioning signal in Stage 1 and a reconstruction target in Stage 2, and is not required at inference.

In standard architectures, backbone visual representations directly condition the action expert. LIT instead routes visual conditioning through a pose-supervised latent interface to an action expert pretrained without images, as detailed below.

B. Stage 1: Spatial-Goal-Conditioned Action Pretraining

Stage 1 trains the action expert from scratch to generate demonstrated action chunks conditioned on language, robot state, and the terminal pose $\mathbf{g}_t$, learning a spatial-goal-conditioned action prior without images.

The frozen backbone processes only the language instruction l and robot state $\mathbf{s}_t$, producing representations $\mathbf{H}_{\ell,t}^{\text{sem}}$ at each of the L coupling layers. A trainable three-layer MLP with GELU activations maps $\mathbf{g}_t$ to goal tokens $\mathbf{G}_t$, which are concatenated with the backbone representations:

$$\begin{aligned} \mathbf{G}_t &= E_\eta(\mathbf{g}_t), \\ \mathbf{C}_{\ell,t} &= [\mathbf{H}_{\ell,t}^{\text{sem}}; \mathbf{G}_t], \quad \ell = 1, \dots, L. \end{aligned} \quad (3)$$

Here, $[\cdot; \cdot]$ denotes token concatenation. $\mathbf{C}_{\ell,t}$ conditions the corresponding action expert layer through the architecture’s native conditioning mechanism.

We retain each framework’s native action-generation objective. For the flow-matching objective used here [36], we sample $\tau \sim \mathcal{U}(0, 1)$ and $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and construct the noisy action chunk and target velocity:

$$\begin{aligned} \tilde{\mathbf{A}}_t^\tau &= (1 - \tau)\epsilon + \tau\mathbf{A}_t, \\ \mathbf{v}_t^* &= \mathbf{A}_t - \epsilon. \end{aligned} \quad (4)$$

Given $\mathbf{C}_{1:L,t} = \{\mathbf{C}_{\ell,t}\}_{\ell=1}^L$, the action expert predicts the velocity and minimizes:

$$\begin{aligned} \hat{\mathbf{v}}_t^\tau &= v_\theta(\tilde{\mathbf{A}}_t^\tau, \tau; \mathbf{C}_{1:L,t}), \\ \mathcal{L}_{\text{prior}} &= \mathbb{E}_{\mathbf{A}_t, \tau, \epsilon} [\|\hat{\mathbf{v}}_t^\tau - \mathbf{v}_t^*\|_2^2]. \end{aligned} \quad (5)$$

Only the action expert parameters θ and SE(3) encoder parameters η are updated; the backbone and its modality encoders remain frozen. The learned action expert parameters initialize Stage 2.

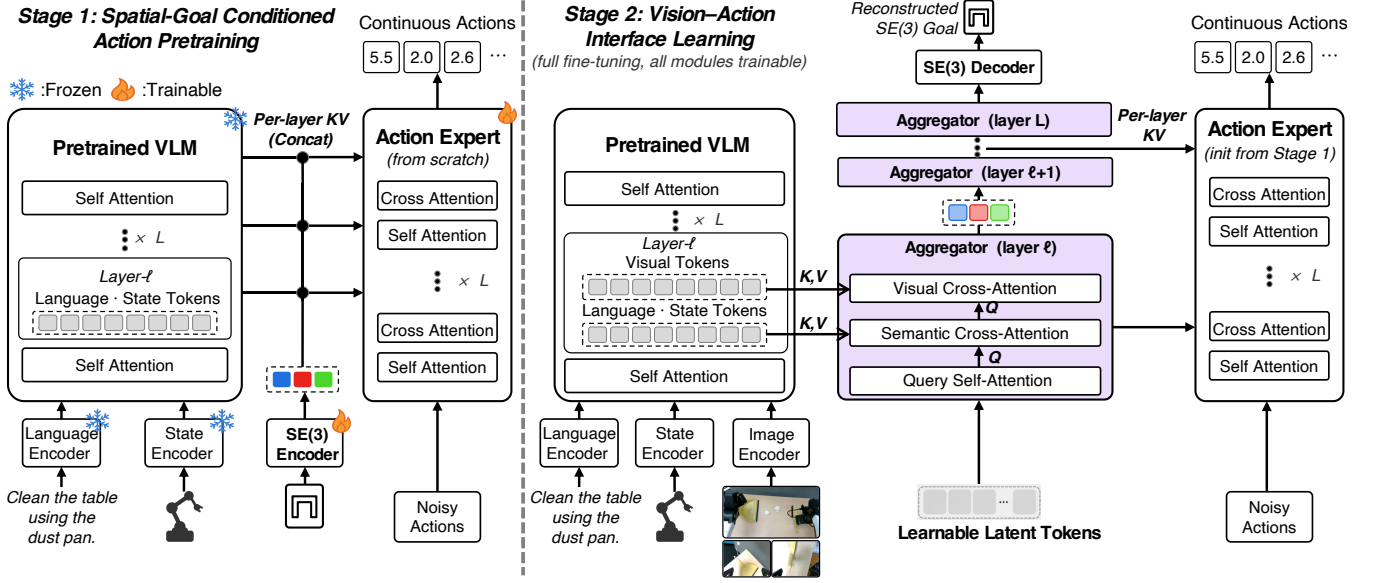


Fig. 2. **Overview of Latent Interface Training (LIT).** Stage 1 learns a spatial-goal-conditioned action prior without images by conditioning the action expert on language, robot state, and each demonstrated action chunk’s terminal SE(3) end-effector pose. Stage 2 introduces visual conditioning through learnable latent tokens that aggregate visual and semantic backbone representations and provide layer-wise conditioning to the pretrained action expert. The tokens are supervised to reconstruct the same terminal pose, linking action-prior learning with visual interface learning. Routing all visual conditioning through these spatially supervised tokens aims to mitigate vision–action shortcuts while preserving task-relevant spatial information.

C. Stage 2: Vision–Action Interface Learning

Stage 2 initializes the action expert from Stage 1 and introduces visual conditioning exclusively through a pose-supervised latent interface.

Let $\mathbf{Z}^0 \in \mathbb{R}^{K \times d}$ denote learnable latent tokens shared across inputs, with $K = 100$ and token dimension d . For each policy input, the interface starts from $\mathbf{Z}_{0,t} = \mathbf{Z}^0$. At coupling layer ℓ , the backbone provides language and state representations $\mathbf{H}_{\ell,t}^{\text{sem}}$ and visual representations $\mathbf{H}_{\ell,t}^{\text{vis}}$. The latent tokens are updated through self-attention, semantic cross-attention, and visual cross-attention:

$$\begin{aligned} \bar{\mathbf{Z}}_{\ell,t} &= \mathbf{Z}_{\ell-1,t} + \text{SA}_{q(\ell)}(\mathbf{Z}_{\ell-1,t}), \\ \tilde{\mathbf{Z}}_{\ell,t} &= \bar{\mathbf{Z}}_{\ell,t} + \text{CA}_{q(\ell)}^{\text{sem}}(\bar{\mathbf{Z}}_{\ell,t}; \mathbf{H}_{\ell,t}^{\text{sem}}), \\ \mathbf{Z}_{\ell,t} &= \tilde{\mathbf{Z}}_{\ell,t} + \text{CA}_{q(\ell)}^{\text{vis}}(\tilde{\mathbf{Z}}_{\ell,t}; \mathbf{H}_{\ell,t}^{\text{vis}}), \quad \ell = 1, \dots, L. \end{aligned} \quad (6)$$

Latent tokens provide the queries in each cross-attention operation, while backbone representations provide the keys and values. For parameter efficiency, every m consecutive coupling layers share interface attention parameters, indexed by $q(\ell) = \lceil \ell/m \rceil$, while accessing their respective backbone representations.

The updated tokens condition the corresponding action expert layers through the architecture’s native conditioning mechanism. Using Eq. 4, the action expert predicts

$$\hat{\mathbf{v}}_t^\tau = v_\theta(\tilde{\mathbf{A}}_t^\tau, \tau; \mathbf{Z}_{1:L,t}), \quad (7)$$

where $\mathbf{Z}_{1:L,t} = \{\mathbf{Z}_{\ell,t}\}_{\ell=1}^L$. The action loss $\mathcal{L}_{\text{act}}$ follows Eq. 5, replacing $\mathbf{C}_{1:L,t}$ with $\mathbf{Z}_{1:L,t}$.

An MLP decoder reconstructs the same terminal goal state

$\mathbf{g}_t$ used to condition Stage 1 from the final latent tokens:

$$\begin{aligned} \hat{\mathbf{g}}_t &= D_\omega(\mathbf{Z}_{L,t}), \\ \mathcal{L}_{\text{pose}} &= \|\hat{\mathbf{g}}_t - \mathbf{g}_t\|_2^2, \\ \mathcal{L}_{\text{stage2}} &= \mathcal{L}_{\text{act}} + \lambda_{\text{pose}} \mathcal{L}_{\text{pose}}. \end{aligned} \quad (8)$$

We set $\lambda_{\text{pose}} = 0.3$. The reconstruction loss is computed in the preprocessed state space and averaged over valid targets, encouraging the interface to retain goal-relevant information. Stage 2 jointly optimizes the backbone and its modality encoders, the action expert, the latent tokens and interface attention modules, and the MLP decoder.

D. Framework Integration and Inference

LIT introduces a latent interface between a pretrained backbone (VLM or video model) and an action expert, making it applicable to VLA and WAM architectures with this structure. The interface provides layer-wise conditioning through each architecture’s native mechanism, while retaining the backbone and action expert architectures, action representations, prediction horizons, and native action-generation objectives.

At inference, the latent interface remains active, while the Stage 1 pose encoder and Stage 2 pose decoder are omitted. The policy requires only visual observations, language, and robot state, and follows each framework’s native action sampling and execution procedure.

IV. EXPERIMENTS

Our experimental evaluation comprises a broad suite of studies designed to assess the effectiveness, robustness, and generality of LIT across different robot foundation model

architectures and deployment settings. Specifically, we investigate LIT along five main dimensions: (i) we examine whether LIT can preserve or improve in-distribution task performance across heterogeneous VLA and WAM architectures; (ii) we evaluate whether LIT improves zero-shot generalization under a diverse set of task-preserving distribution shifts, including changes in camera viewpoints, sensor noise, lighting, background textures, robot initial states, object layouts, and language instructions; (iii) we assess whether these generalization gains transfer to real-world manipulation under changes in lighting, camera configuration, and task-irrelevant distractors; (iv) we analyze whether LIT reduces reliance on spurious vision–action shortcuts by maintaining task-relevant visual attention and responding appropriately to visual and goal interventions; and (v) we study the learning dynamics and key design components of LIT, including Stage 1 spatial-goal-conditioned action-prior learning, pose supervision, and restricting visual conditioning to the latent interface, to determine which factors are responsible for its generalization gains.

A. Experimental Setup

Benchmarks and Evaluation Protocol We evaluate on LIBERO and LIBERO-Plus using task success rate as the metric. We evaluate 40 tasks across the LIBERO-Spatial, LIBERO-Object, LIBERO-Goal, and LIBERO-Long suites [14]. We conduct 50 rollouts per task, totaling 2,000 episodes, to evaluate performance on the original task distribution. LIBERO-Plus evaluates zero-shot generalization under seven task-preserving perturbations: camera viewpoints, sensor noise, robot initial states, language instructions, object layout, lighting conditions, and background textures [7]. We evaluate all 10,030 perturbed instances using one rollout per instance with a fixed evaluation seed and report success rates for each perturbation dimension. Overall denotes the unweighted mean over the seven perturbation dimensions. All models are trained only on the original LIBERO demonstrations and evaluated on LIBERO-Plus without adaptation.

Architectures and Baselines. To evaluate LIT across heterogeneous robot foundation model designs, we integrate it into two VLA architectures, $\pi_{0.5}$ [1] and MolmoAct2 [2], and two WAM architectures, FAST-WAM [3] and ImageWAM [4]. These architectures cover two mechanisms for coupling VLMs with action experts and two different uses of future visual modeling:

- $\pi_{0.5}$: a VLA that couples separate VLM and action expert branches through shared self-attention in a Mixture-of-Transformers architecture.
- MolmoAct2: a VLA whose action expert cross-attends to the corresponding per-layer KV caches of the VLM.
- FAST-WAM: a WAM that uses future-video prediction during training but performs action-only inference without future prediction.
- ImageWAM: a WAM that retains image-editing denoising at inference and conditions its action expert on the resulting KV caches without decoding the target image.

TABLE I
IN-DISTRIBUTION SUCCESS RATES (%) ON LIBERO. BETTER RESULT IN EACH ARCHITECTURE IN BOLD.

Models	Variant	Spatial	Object	Goal	Long	Avg.
$\pi_{0.5}$	Baseline	88.60	93.40	89.20	79.80	87.75
	LIT	90.20	98.80	93.40	84.80	91.80
MolmoAct2	Baseline	93.00	97.80	95.40	87.80	93.50
	LIT	94.60	96.20	95.20	90.40	94.10
FAST-WAM	Baseline	98.20	100.00	97.00	95.20	97.60
	LIT	98.80	99.80	98.40	95.40	98.10
ImageWAM	Baseline	98.40	100.00	97.60	96.40	98.10
	LIT	99.60	99.20	99.20	95.60	98.40

Despite these architectural differences, all four baselines condition action generation on rich visual or world-model representations through their native interaction mechanisms. LIT routes this conditioning through the pose-supervised latent interface while retaining each framework’s backbone and action expert architectures, action representation, and action-generation objective.

Training Details. For each architecture, the baseline and LIT start from the same pretrained backbone and randomly initialized action experts. **We do not fine-tune pretrained VLA or WAM policy checkpoints**, avoiding vision–action dependencies inherited from policy pretraining. Our baseline results are therefore not directly comparable to published results obtained by fine-tuning such checkpoints. Each baseline–LIT pair uses identical training data and matched shared settings, following the architecture’s native configurations. Latent interface dimensions are adapted to the backbone and action expert to match the native conditioning mechanism. Each baseline follows its architecture’s native training budget. LIT allocates the same total number of optimization steps across its two stages, initializing the Stage 2 action expert from Stage 1.

B. Performance Preservation Across Tasks

To determine whether routing visual conditioning through the pose-supervised latent interface preserves action-generation performance, we compare LIT with architecture-matched baselines on the four LIBERO suites. As shown in Table I, LIT preserves or improves the average success rate across all four architectures: from 87.75% to 91.80% for $\pi_{0.5}$, from 93.50% to 94.10% for MolmoAct2, from 97.60% to 98.10% for FAST-WAM, and from 98.10% to 98.40% for ImageWAM. Despite small variations on individual suites, no architecture exhibits an average performance degradation, supporting the effectiveness of the latent interface for in-distribution action generation across heterogeneous VLA and WAM architectures.

C. Generalization Across VLA and WAM Architectures

To evaluate whether LIT improves zero-shot generalization across different robot foundation model families, we compare it with architecture-matched baselines on LIBERO-Plus. As shown in Table II, LIT consistently improves the overall success rate across all evaluated VLA and WAM architectures.

TABLE II

ZERO-SHOT SUCCESS RATES (%) ON LIBERO-PLUS. Δ : CHANGE FROM THE PAIRED BASELINE. OVERALL: MEAN OVER THE SEVEN PERTURBATIONS. BETTER RESULT IN EACH ARCHITECTURE IN BOLD.

Perturbation	$\pi_{0.5}$ (VLA)			MolmoAct2 (VLA)			FAST-WAM (WAM)			ImageWAM (WAM)		
	Base	LIT	Δ	Base	LIT	Δ	Base	LIT	Δ	Base	LIT	Δ
Camera Viewpoints	58.29	80.30	+22.01	39.40	48.41	+9.01	16.40	43.83	+27.43	82.16	84.55	+2.39
Sensor Noise	79.89	91.94	+12.05	49.03	70.77	+21.74	37.70	56.89	+19.19	96.54	97.57	+1.03
Lighting Conditions	82.40	90.11	+7.71	84.86	85.64	+0.78	78.20	83.89	+5.69	97.65	97.99	+0.34
Background Textures	85.32	86.99	+1.67	89.78	94.42	+4.64	53.70	57.22	+3.52	88.26	94.24	+5.98
Robot Initial States	60.71	63.71	+3.00	50.71	59.03	+8.32	44.50	48.86	+4.36	47.81	60.19	+12.38
Object Layout	64.00	73.11	+9.11	55.90	71.61	+15.71	60.70	64.17	+3.47	77.72	83.67	+5.95
Language Instructions	52.15	71.50	+19.35	75.69	73.58	-2.11	68.90	69.55	+0.65	90.97	90.05	-0.92
Overall	68.97	79.67	+10.70	63.62	71.92	+8.30	51.44	60.63	+9.19	83.02	86.89	+3.87

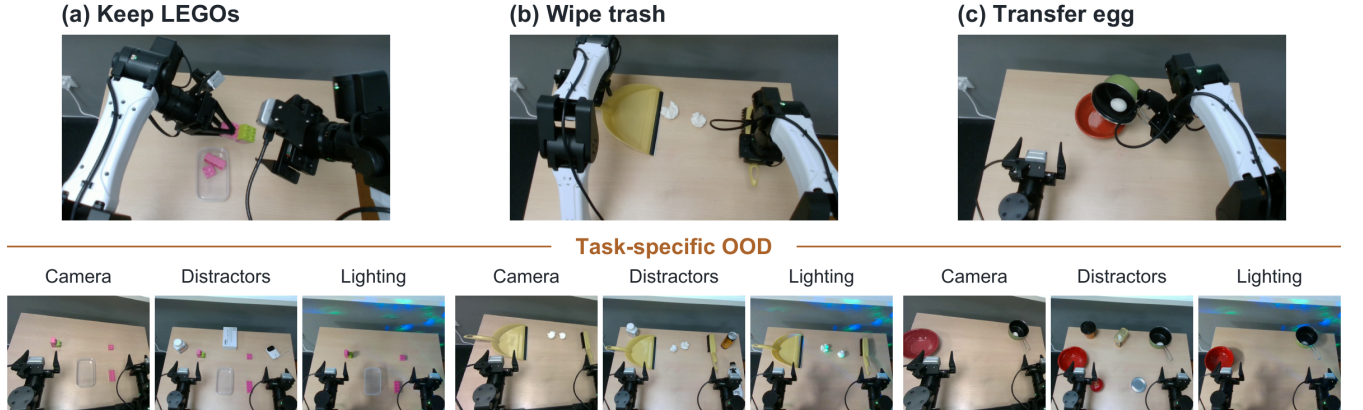


Fig. 3. Real-robot evaluation environments under ID and OOD visual conditions. Rows correspond to the three manipulation tasks, while columns show the in-distribution setting and the three evaluated visual shifts: Lighting OOD, Camera OOD, and Distractors OOD. Camera OOD uses only the top-camera view.

We first examine the two VLA architectures. LIT improves the overall success rate of MolmoAct2 from 63.62% to 71.92%, with substantial gains under Sensor Noise (+21.74 points) and Object Layout (+15.71 points). A similar trend is observed for $\pi_{0.5}$, whose overall success rate increases from 68.97% to 79.67%, with gains of 22.01 points under Camera Viewpoints, 12.05 points under Sensor Noise, and 9.11 points under Object Layout. The improvements under viewpoint and sensing perturbations suggest that LIT reduces reliance on nuisance visual correlations, while the gains under object layout demonstrate that it preserves the spatial information necessary for adaptive action generation. Notably, despite relying on different interaction mechanisms, with layerwise cross attention in MolmoAct2 and shared self attention in $\pi_{0.5}$, both models consistently benefit from LIT. These results demonstrate that LIT generalizes across distinct VLA architectures and visual action interaction mechanisms.

We next examine whether LIT extends beyond VLA architectures to world-action models. LIT improves FAST-WAM from 51.44% to 60.63% overall, with gains of 27.43 points under Camera Viewpoints and 19.19 under Sensor Noise. ImageWAM improves from 83.02% to 86.89%, including gains of 12.38 points under Robot Initial States and 5.95 under Object Layout. FAST-WAM uses future prediction only

during training, whereas ImageWAM retains future-image modeling at inference. Improvements in both settings support LIT’s applicability across distinct world-action modeling pipelines. Overall, LIT improves 26 of the 28 architecture–perturbation comparisons, with no decrease exceeding 2.11 percentage points, supporting its broad effectiveness across the evaluated VLA and WAM architectures.

D. Real-world Evaluations

To evaluate whether the generalization improvements observed in simulation transfer to physical manipulation, we compare LIT with the architecture-matched MolmoAct2 baseline on three real-robot tasks. In *Keep LEGOs*, the robot must detach all blocks and put them into a box. In *Wipe trash*, the robot must clean the table using a dustpan. In *Transfer egg*, the robot must transfer an egg from a pan into a bowl. For each method, we jointly train a single multi-task policy on 300 demonstrations, with 100 demonstrations per task, and evaluate the resulting policy separately on each task.

We evaluate each policy under an in-distribution (ID) setting and three out-of-distribution (OOD) visual conditions, as shown in Fig. 3. In **Lighting OOD**, the scene illumination is changed from that observed during training.

In **Camera OOD**, the policy is evaluated by changing the camera configuration (top-camera only) from the training setting. In **Distractors OOD**, additional task-irrelevant objects are introduced into the scene while leaving the task unchanged. These perturbations test robustness to changes in visual appearance, camera configuration, and scene content, respectively. Each policy is evaluated for 25 rollouts per task under ID conditions (75 rollouts total) and 10 rollouts per task under each OOD condition (90 OOD rollouts total). Fig. 6 reports success rates.

The real-robot results broadly support the generalization trends observed in simulation. Aggregated across the three tasks, LIT improves ID success from 74.7% to 88.0%. More importantly, the improvement persists under all three OOD conditions. LIT improves success from 53.3% to 70.0% under Lighting OOD, from 30.0% to 46.7% under Camera OOD, and from 50.0% to 63.3% under Distractors OOD.

The gains are particularly pronounced on Transfer egg, where LIT improves ID success from 52.0% to 92.0%. Under Lighting OOD and Distractors OOD, success increases from 30.0% to 90.0% and from 20.0% to 90.0%, respectively. On Wipe trash, LIT matches the baseline under ID conditions while substantially improving Camera OOD performance from 10.0% to 80.0%. These results indicate that the generalization benefits of the latent interface extend beyond simulated perturbations to physical changes in the robot’s visual observations.

While the magnitude of the improvement varies across tasks and perturbation types, the overall trend favors LIT. Aggregated across the three tasks, LIT achieves higher success under ID conditions and under each of the three OOD conditions. Together with the simulation results, these findings provide complementary evidence that the latent interface improves robustness to visual distribution shifts in physical manipulation.

E. Analysis and Ablation Studies

Having established LIT’s generalization improvements in Table II, we investigate the behavior, learning dynamics, and design choices underlying these gains. Unless otherwise specified, all analyses and ablations use MolmoAct2. We first examine visual attention and responses to controlled interventions, then analyze learning dynamics across the two training stages, and finally evaluate the contributions of LIT’s components through six ablations.

1) Behavioral Analysis: Stability of action-to-image attention. We examine whether LIT maintains attention to task-relevant regions under distribution shifts by visualizing attention maps for clean and perturbed observations. For the baseline, we use direct action-to-visual attention. For LIT, we compose attention from the action expert to the latent tokens with attention from the latent tokens to visual representations. Figure 4 compares the baseline (top row) and LIT (bottom row). Baseline attention shifts across perturbations, whereas LIT’s attention remains more concentrated on task-relevant robot-object regions. These qualitative observations suggest more stable visual attention during action generation.

Responses to visual and goal interventions. We probe reliance on vision-action shortcuts through two types of counterfactual interventions (Fig. 5). For *task-preserving visual interventions*, we add a distractor or blur the image while keeping the instruction, spatial goal, and robot state fixed (left and middle columns). The baseline trajectories deviate substantially from their clean counterparts, whereas LIT’s remain close. For the *goal-changing intervention*, we change the instructed goal while keeping the visual scene and robot state fixed (right column). The baseline continues toward the original goal, consistent with reliance on unchanged visual cues, whereas LIT redirects toward the new goal. Together, LIT’s robustness to task-irrelevant visual changes and responsiveness to goal changes provide evidence consistent with reduced reliance on vision-action shortcuts.

2) Learning Dynamics: We examine whether the training dynamics align with the intended roles of action-prior learning and visual-interface learning (Fig. 7). In Stage 1, the action loss decreases rapidly to 0.029 after 10K steps, indicating that the action expert learns goal-directed action generation without images. In Stage 2, the pose-reconstruction loss reaches 0.003 after 20K steps, showing that the demonstrated spatial goals are recoverable from the learned interface on training data. LIT also achieves a lower action loss in 20K visual training steps than the baseline reaches in 30K steps, with both methods using the same total training budget. These dynamics support the intended learning roles of the two stages and complement the behavioral analyses and generalization results.

3) Component Ablations: We evaluate six ablations in Table III. We first examine the contributions of action pre-training, pose supervision, and the restricted visual pathway by removing individual components from LIT. We then test whether latent-token aggregation, staged training, or pose supervision alone can account for its gains.

Contribution of the action prior. We skip Stage 1 and randomly initialize the action expert, while retaining the Stage 2 latent interface and pose supervision (*LIT w/o Stage 1*). OOD success decreases from 71.92% to 68.23%. This 3.69-percentage-point decrease supports the benefit of spatial-goal-conditioned action pretraining for subsequent visual policy learning.

Contribution of pose supervision. We remove the Stage 2 pose-reconstruction loss while retaining all other components (*LIT w/o pose supervision*). OOD success decreases from 71.92% to 68.86%, a reduction of 3.06 percentage points. The largest decreases occur under Sensor Noise (from 70.77% to 60.02%, 10.75 percentage points) and Background Textures (from 94.42% to 88.38%, 6.04 points). These results are consistent with spatial-goal supervision improving robustness to task-irrelevant visual perturbations.

Contribution of the restricted visual pathway. We allow the action expert to directly access backbone visual representations while retaining all LIT components (*LIT w/ direct visual access*). Although LIBERO success remains comparable to LIT (94.25% vs. 94.10%), LIBERO-Plus success decreases from 71.92% to 67.74%. This 4.18-percentage-

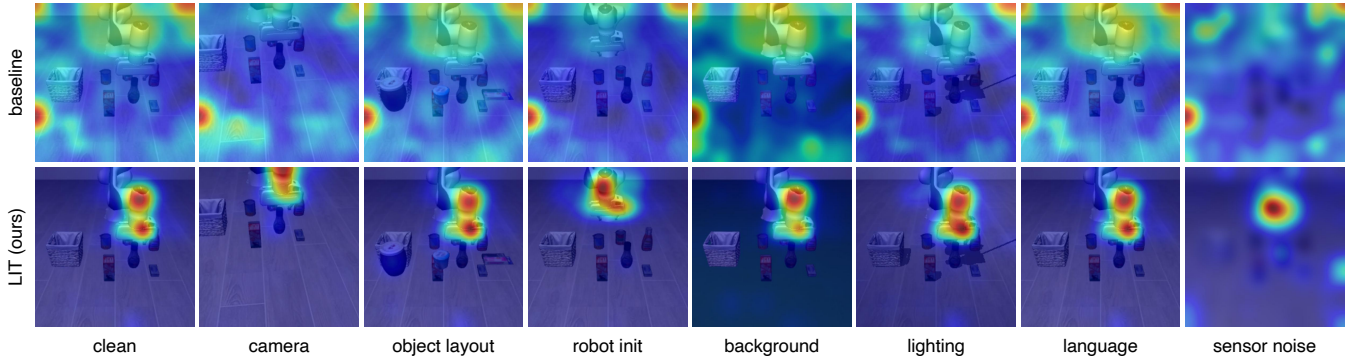


Fig. 4. Effective action-to-image attention under distribution shifts. Baseline attention varies substantially across perturbations, whereas LIT remains consistently aligned with task-relevant robot-object regions.

TABLE III

ABLATION STUDY OF LIT ON MOLMOACT2. TASK SUCCESS RATES (%) ON LIBERO (ID) AND LIBERO-PLUS (OOD). OVERALL DENOTES AGGREGATE OOD SUCCESS. HIGHER IS BETTER. COMPONENT ABLATIONS REMOVE INDIVIDUAL ELEMENTS FROM LIT; ALTERNATIVE DESIGNS TEST SIMPLER EXPLANATIONS FOR ITS GAINS. THE BEST AND SECOND-BEST RESULTS IN EACH COLUMN ARE SHOWN IN **BOLD** AND UNDERLINED, RESPECTIVELY.

Variant	ID	OOD: LIBERO-Plus							
	LIBERO Avg.	Camera Viewpoints	Sensor Noise	Lighting Conditions	Background Textures	Robot Initial States	Object Layout	Language Instructions	Overall
MolmoAct2	93.50	39.40	49.03	84.86	89.78	50.71	55.90	75.69	63.62
<i>Component ablations</i>									
LIT w/o Stage 1	93.70	<u>46.15</u>	54.65	87.13	86.34	63.10	66.43	<u>73.80</u>	68.23
LIT w/o pose supervision	93.50	42.78	<u>60.02</u>	91.07	88.38	57.81	<u>68.85</u>	73.11	<u>68.86</u>
LIT w/ direct visual access	94.25	42.96	55.28	88.18	89.22	<u>62.49</u>	63.74	72.33	67.74
<i>Alternative designs</i>									
LIT w/o Stage 1 & pose supervision	93.45	39.84	49.34	87.22	89.63	53.00	68.20	72.67	65.70
LA4VLA-inspired staged training [12]	93.75	43.28	52.09	85.17	88.08	55.16	60.98	73.46	65.46
Baseline w/ pose supervision	93.20	40.32	51.30	<u>89.00</u>	<u>90.13</u>	55.45	58.37	73.56	65.45
LIT	<u>94.10</u>	48.41	70.77	85.64	94.42	59.03	71.61	73.58	71.92

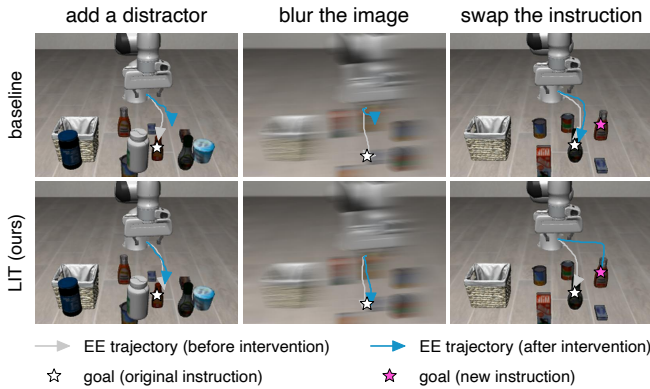


Fig. 5. Counterfactual trajectory analysis. LIT remains stable under task-preserving visual interventions and redirects its trajectory when the specified goal changes.

point decrease supports the importance of routing visual information exclusively through the pose-supervised latent interface for generalization.

Latent-token aggregation alone. To assess whether latent-

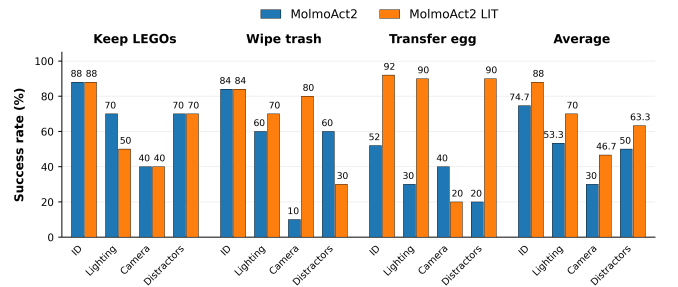


Fig. 6. Real-robot task success under in-distribution (ID) and out-of-distribution (OOD) visual conditions. Camera OOD uses only the top-camera view. Bars report success rates (%).

token aggregation accounts for LIT’s gains, we jointly remove Stage 1 and pose supervision (*LIT w/o Stage 1 & pose supervision*). Learnable tokens still aggregate backbone representations to condition the action expert, as in query-based interfaces such as VLA-Adapter [37], but are trained solely through the action objective. OOD success reaches 65.70%, only 2.08 percentage points above the baseline

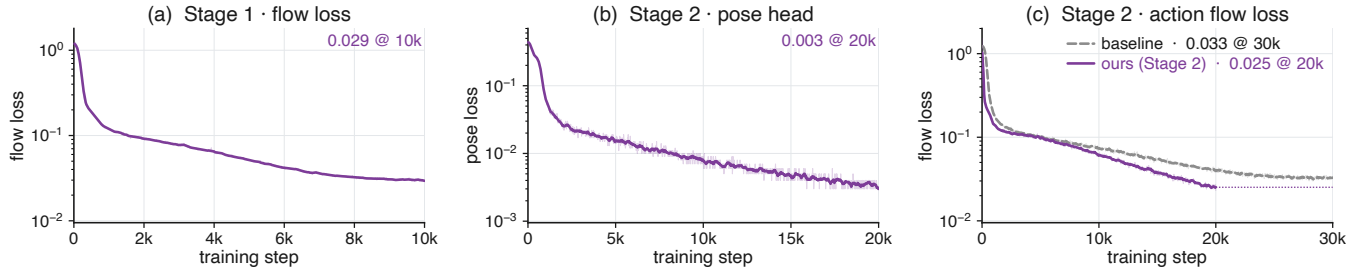


Fig. 7. **Training dynamics on MolmoAct2.** (a) Stage 1 action flow-matching loss. (b) Stage 2 pose-reconstruction loss. (c) Action flow-matching loss during visual policy training. LIT uses 10K Stage 1 and 20K Stage 2 steps, matching the baseline’s 30K-step total budget. LIT’s Stage 2 action loss reaches a lower value than the baseline’s final loss. Panel (c) counts visual training steps; the 0.63 $\times$ ratio compares losses at 20K visual training steps.

and 6.22 points below LIT. This result supports combining the latent interface with action pretraining and spatial-goal supervision.

Staged training alone. To assess whether two-stage training accounts for LIT’s gains, we evaluate an LA4VLA-inspired baseline [12] (*LA4VLA-inspired staged training*). It first predicts actions from language and robot state without images, then conditions the action expert on backbone visual representations. OOD success increases from 63.62% for MolmoAct2 to 65.46%, but remains 6.46 percentage points below LIT. Thus, the tested staged-training alternative does not account for LIT’s full gains.

Pose supervision alone. We add the same pose-reconstruction objective to the baseline while retaining its original visual conditioning architecture (*Baseline w/ pose supervision*). An MLP decoder reconstructs the terminal goal state from the baseline’s final-layer backbone features. OOD success increases from 63.62% to 65.45%, but remains 6.47 percentage points below LIT. This modest improvement suggests that pose supervision alone does not explain LIT’s gains, supporting its integration with the action prior and restricted latent interface.

V. LIMITATIONS AND CONCLUSION

While our experiments demonstrate improved robustness to visual variations, evaluation at larger real-world scales remains open. Future work will explore scaling LIT to larger and more diverse robot demonstration datasets and extending its evaluation to a broader range of tasks and environments.

We presented Latent Interface Training (LIT), a framework-agnostic two-stage strategy for mitigating vision-action shortcuts in robot foundation models. LIT combines image-free, spatial-goal-conditioned action pretraining with a pose-supervised latent interface as the action expert’s only visual conditioning pathway. Across four VLA and WAM architectures, LIT preserves in-distribution performance on LIBERO while improving zero-shot generalization on LIBERO-Plus. Real-world experiments further show improved robustness to unseen visual variations. These findings support LIT’s joint design for reducing shortcut reliance while retaining the spatial information needed for action generation.

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